File permissions in Linux

Project description

The following Linux commands described below were used to check file permissions of the `projects` directory for the `researcher2` user, including hidden files, to make sure they align with the authorisation given by the organisation. Checking and updating these permissions will help keep the organisation's system secure. To complete this task, I performed the following tasks:

Check file and directory details

The following code demonstrates how I used Linux commands to determine the existing permissions set for a specific directory in the file system.

```
researcher2@b1ffb635cb36:~$ cd projects
```

```
researcher2@b1ffb635cb36:~/projects$ ls -la
```

I used `cd projects` command to first navigate to the `projects` directory, then used the `ls -la` command to display permissions for all files, including hidden files, in the project's directory.

Output:

```
researcher2@b1ffb635cb36:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jan 14 19:06 .
drwxr-xr-x 3 researcher2 research_team 4096 Jan 14 19:22 ..
-rw--w---- 1 researcher2 research_team   46 Jan 14 19:06 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jan 14 19:06 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Jan 14 19:06 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Jan 14 19:06 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jan 14 19:06 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jan 14 19:06 project_t.txt
researcher2@b1ffb635cb36:~/project
```

The output of my command indicates that there is one directory named `drafts`, one hidden file named `.project_x.txt`, and five other project files. The 10-character string in the first column represents the permissions set on each file or directory.

Describe the permissions string

```
-rw-rw-rw- 1 researcher2 research_team   46 Jan 14 19:06 project_k.txt
```

The 10-character string is used to describe the permissions for the user, group and other owner types of the file or directory. For example, the screenshot above is the output from the `ls -la` command, and it is one example of what a 10-character string could be for the file `project_k.txt` with a 10-character string `-rw-rw-rw-`.

The **1st character** indicates the file type, which is either `d` or hyphen (`-`). If it's a `d`, it's a directory. If it's a hyphen (`-`), it's a regular file. In the screenshot above, the first character is a hyphen, which means it's a regular file.

The **2nd-4th characters** indicate the read (`r`), write (`w`), and execute (`x`) permissions for the user. When one of these characters is a hyphen (`-`) instead, it indicates that this permission is not granted to the user. In the screenshot above, the user has read (`r`) and write (`w`) permissions.

The **5th-7th characters** indicate the read (`r`), write (`w`), and execute (`x`) permissions of the group. When one of these characters is a hyphen (-) instead, it indicates that this permission is not granted to the group. In the screenshot above, the group has read (r) and write (w) permissions.

The **8th-10th characters** indicate the read (`r`), write (`w`), and execute (`x`) permissions for other. When one of these characters is a hyphen (-) instead, it indicates that this permission is not granted for other. In the screenshot above, the other user type has read and write permissions. In the screenshot above, the group has read (`r`) and write (`w`) permissions.

For example, the file permissions for project_t.txt are -rw-rw-r--. Since the first character is a hyphen (-), this indicates that project_t.txt is a file, not a directory. The second, fifth, and eighth characters are all r, which indicates that user, group, and other all have read permissions. The third and sixth characters are w, which indicates that only the user and group have write permissions. No one has execute permissions for project_t.txt.

Change file permissions

```
researcher2@27227ebdad7f:~/projects$ chmod o-w project_k.txt
researcher2@27227ebdad7f:~/projects$ ls -l
```

From the output shown in 'Check file and directory details' it shows that **project_k.txt** has permissions allowing other users to write. The write permission must be removed because the organisation doesn't allow others to have write access to any files.
Therefore, the **chmod o-w project_k.txt** command was used to remove the write permission from other users, as shown in the screenshot above. **chmod** is used to modify the permissions

of files and directories. The first argument is used to describe what permission you want to modify, and the second argument specifies the file or directory. In the example in the screenshot above, I removed write permissions from other for the `project_k.txt` file. After this, I used `ls -la` to review the updates I made.

I then used the `ls -l` command to confirm that the write permission was removed, as shown in the output below. In the 10 characters, you can see there are no write permissions granted for other in project_k.txt.

Output:

```
drwx--x--- 2 researcher2 research_team 4096 Jan 14 17:32 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Jan 14 17:32 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Jan 14 17:32 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jan 14 17:32 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jan 14 17:32 project_t.txt
```

Change file permissions on a hidden file

```
researcher2@27227ebdad7f:~/projects$ chmod u-w,g-w,g+r .project_x.txt
```

```
researcher2@27227ebdad7f:~/projects$ ls -la
```

From the output shown in 'Check file and directory details', it shows you `.project_x.txt` has write permissions for both the user and the group and read permissions for the user, but not for the group. The `.project_x.txt` is a hidden file due to the . at the start of the file name.

However, the organisation does not want anyone to have write access, but they want the user and group to have read access.

So the permissions must be changed to meet the requirements. As a result, I used the command `chmod u-w,g-w,g+r .project_x.txt.` `u-w` and `g-w` are used to take away the write permissions from the user and group. `g+r` is used to add the read permission to the group.

Then I used `ls -la` to show the permissions for all files, including hidden, to confirm that the changes to the `.project_x.txt` file have been made. As you can see from the output, the 10 characters associated with `.project_x.txt` only show read permissions for the user and group.

Output:

```
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jan 14 17:32 .
drwxr-xr-x 3 researcher2 research_team 4096 Jan 14 18:08 ..
-r--r----- 1 researcher2 research_team   46 Jan 14 17:32 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jan 14 17:32 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Jan 14 17:32 project_k.txt
-rw------- 1 researcher2 research_team   46 Jan 14 17:32 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jan 14 17:32 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jan 14 17:32 project_t.txt
```

Change directory permissions

```
researcher2@27227ebdad7f:~/projects$ chmod g-x drafts
researcher2@27227ebdad7f:~/projects$ ls -l
```

From the output shown in the previous section, you know that the `drafts` directory has execute permissions for the group. The organisation only wants the `researcher2` user to have access to the `drafts` directory and its contents. This means no one should have permissions apart from the user, `researcher2`. As a result, the `chmod g-x drafts` command was used to remove the execute permission from the group.

The `ls -l` command was then used to confirm that the changes were made. The 10 characters associated with `drafts` indicate that there are only permissions for the user (read, write and execute), which complies with the requirement.

Output:

```
researcher2@27227ebdad7f:~/projects$ ls -l
total 20
drwx------ 2 researcher2 research_team 4096 Jan 14 17:32 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Jan 14 17:32 project_k.txt
-rw------- 1 researcher2 research_team   46 Jan 14 17:32 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jan 14 17:32 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jan 14 17:32 project_t.txt
```

Summary

I changed multiple permissions to match the level of authorisation my organisation wanted for files and directories in the `projects` directory. The first step was to use `ls -la` to check the directory permissions. This informed my decisions in the following steps. I then used the `chmod` command multiple times to change the permissions on files and directories.